

Uwaise Ibna Islam

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EDUCATION

University of Florida PhD in Computer Science (Bioinformatics) Advisor: Dr. Christina Boucher · Research: scalable algorithms for population genomics, variant analysis, genomic data structures	Expected May 2028 GPA: 4.00
Florida Atlantic University MS in Computer Science Presidential Fellowship; Academic Excellence Award (2023, 2024)	Aug 2022 – May 2024 GPA: 3.97
Chittagong University of Engineering & Technology BS in Computer Science & Engineering	Feb 2016 – Jun 2021

TECHNICAL SKILLS

Languages	Python, R, C++, Bash/Shell, SQL
NGS & Bioinformatics	WGS/WES variant calling, VCF/BCF, FASTA/FASTQ, BAM/CRAM, HTSlib, bcftools, samtools, GATK, PLINK, SHAPEIT, BEDtools, Bioconductor, sequence alignment, copy-number analysis, allele-frequency filtering
Pipelines & Infra	Nextflow, Snakemake, SLURM/HPC, Docker, CI/CD, Google Cloud Platform, AWS
ML & Data Science	PyTorch, TensorFlow, scikit-learn, deep learning, feature engineering, NumPy, pandas
Systems	Linux/UNIX, Git/GitHub, multithreading (OpenMP), memory profiling (perf, Valgrind)

RESEARCH EXPERIENCE

Graduate Research Assistant · University of Florida, Boucher Lab	Jul 2025 – Present
- Designed and implemented PBML, a C++17 tool for scalable haplotype matching on WGS panels from the 1000 Genomes Project (5,008 haplotypes, 1–6M variants/chr) and BIG Initiative biobank (11,000 haplotypes). Achieved 5× faster queries and 44× lower peak memory (4 GB vs 176 GB) compared to SOTA methods; scaled to 50K haplotypes while remaining 4× faster with 2.2× less memory than SOTA. - Built OpenMP multi-threaded query engine achieving 8.2× speedup at 16 threads; length-threshold filtering reduced query time by 74% and output size by 98% while retaining 97% site coverage — enabling practical population-scale IBD detection. - Developing a novel IBD segment detection tool benchmarked on 1KGP and UK Biobank-scale datasets, achieving 2.8× faster runtime and 5× lower peak memory than the current SOTA method (hap-ibd) on chromosome-scale WGS data. - Built automated SLURM pipelines for large-scale WGS phasing and variant calling (SHAPEIT5); designed job scheduling, containerized environments, and error recovery for batch HPC processing.	
Graduate Research Assistant · Florida Atlantic University, Assis Group	Aug 2022 – May 2025
- Developed end-to-end deep learning pipeline for cross-species genotype–phenotype prediction with custom feature engineering on multi-species genomic alignments (59 species). Near-perfect cross-validated accuracy across multiple phenotype tasks. - Authored R/Bioconductor package with parallel genomic analysis workflows and integrated neural network inference; achieved 7× speedup through optimized preprocessing. Published in <i>Briefings in Bioinformatics</i> (2025).	

PUBLICATIONS

- Islam U.I.**, Cozzi D., Gagie T., Bonizzoni P., Varki R., Colonna E., Garrison E., Boucher C. Scaling the PBWT for Long-Range Shared Ancestry Detection in Large Haplotype Panels. Submitted to *Bioinformatics* (under review).
- Islam U.I.**, Campelo dos Santos A.L., Kanjilal R., Assis R. Learning genotype-phenotype associations from gaps in multi-species sequence alignments. *Briefings in Bioinformatics*, 2025.

INDUSTRY EXPERIENCE

Software QA Engineer · Enosis Solutions	Aug 2021 – Jul 2022
Automated end-to-end testing with Python/Selenium (63% coverage, 40% fewer regression bugs). Performed API performance testing and collaborated cross-functionally in agile sprints to achieve 82% response time improvement.	

HONORS & CERTIFICATIONS

FAU Presidential Fellowship (2022–2024) · Academic Excellence Award (2023, 2024) · IBM Data Science Professional Certificate · ML Specialization (Coursera)